

Rhutam Mahajan

College Park, MD | +1 (227) 275-1067 | rhutammahajan@gmail.com | linkedin.com/in/rhutammahajan
github.com/Rhutam03 | rhutammahajan.com

EDUCATION

University of Maryland, College Park

M.S. in Data Science | GPA: 3.95/4.0

Aug 2025 – May 2027

College Park, MD

Savitribai Phule Pune University

B.E. in Computer Engineering, Honors in Data Science | GPA: 8.5/10

Dec 2021 – Jun 2025

Pune, India

TECHNICAL SKILLS

Languages & Core Libraries: Python, SQL, Java, JavaScript, TypeScript, Pandas, NumPy, SciPy

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, computer vision, NLP, few-shot and multimodal learning, time-series forecasting

Generative AI & LLMs: OpenAI Agents SDK, Model Context Protocol (FastMCP), LangChain, LangGraph, ChromaDB, RAG, tool calling

Big Data, Cloud & Tools: Apache Spark, PySpark, Spark MLlib, Neo4j, MySQL, SQLite, AWS (ECR, S3, CloudFront, App Runner), Docker, FastAPI, Flask, REST APIs, Git, Tableau, Power BI

EXPERIENCE

InternPe

AI/ML Intern

Feb 2024 – Mar 2024

Remote

- Built four ML pipelines spanning binary, multiclass, and regression tasks with scikit-learn, TensorFlow, and Pandas.
- Lifted match-outcome classifier accuracy from 71.11% to 94.87% by comparing Logistic Regression and Random Forest after engineering live-context features over 149,578 IPL match records.
- Trained models on datasets up to 179,078 records: 96.49% accuracy on a breast-cancer neural net and R^2 0.9210 on car-price regression.

Gamaka AI

Junior Data Scientist

Dec 2023 – Mar 2024

Pune, India

- Audited structured datasets with SQL and statistical checks, flagging missing values, duplicates, inconsistent categories, and outliers for reliable downstream reporting.
- Wrote SQL queries across seven companies' financial records (2010–2012) to analyze revenue, expenses, profit, EPS, and year-over-year trends.

PROJECTS

Agentic AI Systems with MCP, RAG, and Evaluator Loops

OpenAI Agents SDK, MCP (FastMCP), LangGraph, LangChain, ChromaDB

Apr 2026

- Built async multi-agent workflows in Python spanning research, tool calling, memory, retrieval, and evaluation over MCP servers.
- Wrote a custom FastMCP server exposing tools to read state, execute actions, update strategy, and serve resources to agents.
- Built a RAG chatbot with document ingestion and Chroma vector search, using OpenAI for generation and Gemini for evaluation.

Spark Readmission Pipeline & Graph Data Engineering

PySpark, Spark MLlib, Neo4j

Apr 2026

- Built a PySpark cleaning and modeling pipeline on 12,465 readmission records, handling missing values, mismatches, and outliers.
- Engineered eight numeric and eight categorical features through Spark ML pipelines (StringIndexer, OneHotEncoder, VectorAssembler) and trained a Spark MLlib Random Forest (100 trees, max depth 8) on an 80/20 split.
- Scored 60.57% accuracy, 58.29% F1, 61.60% precision, and 0.636 AUC on held-out data.

Multimodal Medical Triage

PyTorch, FastAPI, Docker, GitHub Actions, AWS, React TypeScript

Dec 2025 – Mar 2026

- Built a multimodal triage model that fuses lesion images (ResNet-18 encoder) with clinical text (128-dim embedding encoder) through a shared classifier head to predict Low, Medium, or High risk.
- Trained on 5,067 cases with an 80/20 stratified split using class-weighted cross-entropy and AdamW, reaching 80.13% accuracy and 77.32% macro F1.
- Built a FastAPI inference service with a React + TypeScript frontend, Dockerized the backend, and deployed to AWS (ECR, App Runner, S3) with GitHub Actions CI/CD.

Plant Leaf Disease Classification via Few-Shot Learning

EfficientNet-B0, Prototypical Networks, PyTorch

Aug 2024 – May 2025

- Built a two-level few-shot classifier on EfficientNet-B0, collapsing 38 raw classes into 12 disease categories.
- Trained a 384-dim EfficientProtoNet on balanced few-shot splits (240 images) and evaluated on 13,135 held-out PlantVillage images: 97.77% accuracy, 95.46% macro precision, and 95.83% macro F1.

PUBLICATIONS

Plant Leaf Disease Multilevel Classification Using Few-Shot Learning

International Journal of Research and Analytical Reviews

May 2025